Event-Driven Healthcare Regime Simulation

Model Description

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1. Purpose

This document describes the computational simulation model supporting the framework introduced in:

Event-Driven Stochastic Regime Shifts in Adaptive Healthcare Systems: Modelling Overload, Recovery, and Policy Response

The purpose of the simulation framework is to investigate how adaptive healthcare systems respond to stochastic demand variations, external shocks, and delayed policy interventions.

The model focuses on the emergence of operational regime transitions rather than only steady-state performance metrics.

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2. System Model

The healthcare system is represented as an adaptive queueing system where incoming demand competes against available service capacity.

The baseline system follows a queueing-inspired representation with:

λ : patient arrival rate
μ : service rate per resource/server
c : available system capacity

The effective processing capacity is represented as:

$\text{Capacity} = \mu \times c$

System pressure emerges when incoming demand exceeds the available adaptive response capability.

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3. Regime States

The simulation represents the system through three operational regimes.

NORMAL Regime

The system operates within available capacity limits.

Characteristics:

- stable queue evolution
- controlled waiting times
- sufficient resource availability

OVERLOAD Regime

An overload regime occurs when increased demand or external shocks push the system beyond its operational capacity.

Characteristics:

- increasing queue accumulation
- increased waiting time
- resource saturation

RECOVERY Regime

Recovery represents the adaptive phase following intervention.

Characteristics:

- capacity adjustment
- queue reduction
- return toward stable operation

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4. Event-Driven Dynamics

The model follows an event-driven simulation approach.

Events include:

- demand surge events
- external disruption events
- regime transition events
- adaptive policy actions

External shocks modify the demand profile dynamically.

Policy response is introduced after an adaptation delay:

τ : policy response delay

The delay parameter allows investigation of how intervention timing affects overload duration and recovery efficiency.

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5. Simulation Methodology

The simulation generates discrete-time system evolution traces.

For each scenario, the framework records:

- system state
- queue evolution
- regime transitions
- intervention timing
- recovery behaviour

Multiple scenarios can be evaluated by modifying:

- demand intensity
- service capacity
- shock duration
- response delay

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6. Simulation Outputs

Generated outputs include:

- event logs
- timeline datasets
- summary performance indicators
- visualization figures

Key evaluation metrics:

- maximum queue length

- overload duration
- recovery time
- waiting time
- throughput

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7. Reproducibility

The Python implementation supporting this model is publicly available.

Repository:

https://github.com/dimpapak/event-driven-healthcare-regime-simulation

The repository contains:

- simulation source code
- configuration parameters
- generated example datasets
- figure generation scripts
- documentation

This supports transparency, reproducibility, and future extensions of the proposed framework.

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